

Olist Marketplace Analysis

Order Growth, Delivery Performance, and the Drivers of Customer Satisfaction

Analysis Report

Hackathon Data Analysis Submission

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Data Source: Olist Brazilian E-Commerce Public Dataset

Scope: 99,441 orders across 9 relational tables (Sept. 2016 – Oct. 2018)

Method: SQL-based relational analysis with Python statistical validation

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1. Problem Understanding

Olist is a Brazilian e-commerce marketplace that connects small and medium-sized sellers to major retail channels, handling logistics, payments, and customer reviews on their behalf. The dataset underlying this analysis covers **99,441 orders** placed between September 2016 and October 2018, spanning customers, sellers, products, order items, payments, reviews, and geolocation records across nine relational tables.

The objective of this analysis is to convert this raw operational data into a business-facing understanding of marketplace health, and specifically to answer the following core questions:

1. **Marketplace Performance Over Time** – How have order volume, revenue, and review scores trended, and do they tell a consistent story?
2. **Delivery Performance and Customer Satisfaction** – How does delivery timing relative to the estimated delivery date relate to review scores?
3. **Seller and Geographic Patterns** – How does seller–customer distance relate to freight cost, delivery performance, and satisfaction?
4. **Product Category Performance** – How do order volume, price, and review scores differ across product categories?
5. **Payment Behavior** – How do payment type and installment choices relate to order value and customer experience?
6. **Root Cause Analysis** – Which of the above factors are primary versus secondary drivers of low review scores?

The end goal is a small set of statistically grounded, actionable recommendations for the operations and customer-experience teams – not an exhaustive academic exploration of the dataset.

2. Analytical Approach

2.1. Data Auditing and Cleaning

Before analysis, all nine source files were audited for the three classes of issues that typically corrupt relational analysis: missing primary keys, orphaned foreign keys, and duplicate rows.

- **Primary keys:** No nulls were found in any primary-key column across all eight keyed tables.
- **Foreign-key integrity:** All child–parent relationships (orders→customers, order_items→orders/products/sellers, payments→orders, reviews→orders) were fully intact with zero orphaned rows. The one exception was `products.product_category_name` → `category_translation`, where 13 product rows referenced 2 category names (`pc_gamer`, `portateis_cozinha_e_preparadores_de_alimentos`) absent from the translation table. This was handled with a `COALESCE` fallback to the original Portuguese name during SQL aggregation rather than dropping the rows.
- **Duplicate rows:** No exact duplicate rows were found in any table except `geolocation`, which was de-duplicated before loading (it is a many-to-one zip-code lookup table, not an entity table, so duplicate coordinate rows would otherwise inflate join cardinality). Separately, 814 `review_id` values appeared attached to more than one `order_id` – a genuine characteristic of the source data (a customer’s review can apply to more than one order) rather than a data error, confirmed by verifying that no (`review_id`, `order_id`) pair was itself duplicated.
- **Known data-quality characteristics:** Consistent with the dataset’s documentation, `order_approved_at` (160 orders), `order_delivered_carrier_date` (1,783 orders), and

`order_delivered_customer_date` (2,965 orders) contain nulls corresponding to orders that never completed the normal fulfillment flow; 610 products lack category and listing meta-data; and the large majority of reviews (88,285 of 99,224) have no free-text comment. These were not treated as errors – delivery-timing analysis was deliberately restricted to orders with both a delivery date and an estimated delivery date, and category analysis excludes the small “Unknown” bucket from ranked comparisons.

2.2. Data Modeling

The cleaned CSVs were loaded into a normalized **SQLite database** with an explicit schema: nine tables with declared primary keys and foreign-key constraints (customers, sellers, products, category_translation, orders, order_items, order_payments, order_reviews, geolocation). Referential integrity was verified after load using `PRAGMA foreign_key_check` (zero violations) and `PRAGMA integrity_check` (ok). This normalized structure allowed each business question to be answered with a purpose-built SQL query rather than repeatedly re-joining flat files.

2.3. Query and Statistical Methodology

- **Aggregation logic:** Revenue and order value were computed as `price + freight_value` summed per order (via CTEs) to correctly handle multi-item orders (13,984 orders have more than one item) and multi-payment orders (4,446 orders have more than one payment record) without double-counting or under-counting.
- **Delivery delay:** Defined as `julianday(delivered_customer_date) - julianday(estimated_delivery_date)`, so positive values indicate a late delivery and negative values an early one. Computed only for orders with both dates populated.
- **Geographic distance:** Zip-code prefixes were mapped to mean latitude/longitude from the geolocation table (aggregated first to avoid row multiplication), then seller–customer great-circle distance was computed with the **Haversine formula**.
- **Statistical association:** Both **Pearson** (linear) and **Spearman** (rank/monotonic) correlation coefficients were computed for the delay–review and distance–freight relationships, since review scores are ordinal and several relationships are visibly non-linear.
- **Tooling:** SQLite for relational aggregation; Python (pandas, NumPy) for statistical summaries and binning; Matplotlib/Seaborn for static exploratory charts; Plotly for an interactive executive dashboard reusing the same pre-computed tables (no analysis was duplicated between the exploratory and dashboard sections of the notebook).

This report presents only visualizations and figures that were already produced in the source analysis; no new modeling or computation was performed to produce it.

3. Key Insights Addressing the Core Questions

3.1. Marketplace Performance Over Time

Monthly order volume grew from a low base in late 2016 to a sustained 6,000–7,000 orders/month through most of 2018, with revenue tracking order volume almost exactly – growth in revenue reflects *more orders*, not larger baskets. Review scores, by contrast, stayed comparatively flat around 4.0–4.2 throughout the growth period. Growth visibly slowed toward the end of the observed window.

Key finding. Marketplace scale increased substantially without a corresponding long-term deterioration in average customer satisfaction – the two metrics tell genuinely different stories.

3.2. Delivery Performance and Customer Satisfaction

At the individual-order level, delivery delay and review score are only weakly correlated (Pearson $r = -0.266$, Spearman $\rho = -0.176$). However, grouping orders by delay bucket reveals a sharp, non-linear pattern: average review score is essentially flat (4.15–4.31) for every group delivered on time or early, and falls off a cliff once deliveries run late – dropping to roughly 1.6–2.3 for deliveries more than 4 days late.

Key finding. Delivery lateness behaves like a threshold effect, not a smooth linear one: a modest global correlation coefficient understates a very large practical effect concentrated in the “late” tail.

3.3. Seller and Geographic Patterns

Seller–customer distance is moderately correlated with freight cost at the individual line-item level (Pearson $r = 0.389$), and the grouped average is clearly and monotonically increasing – from roughly \$12 for shipments under 100 km to \$36 for shipments of 2,000–4,000 km. Despite this, distance shows essentially no relationship with lateness: average delivery delay is *negative* (i.e., early) across every distance band, meaning Olist’s estimated-delivery-date logic already absorbs the extra travel time for distant shipments. The relationship with review score is real but modest, declining from about 4.28 (under 100 km) to 3.98 (2,000–4,000 km); the final bucket (over 4,000 km) drops further to roughly 3.0 but is based on only 4 orders and should not be treated as reliable.

Key finding. Geography is primarily a *cost* problem, not an experience problem – Olist’s logistics network appears to compensate operationally for distance.

3.4. Product Category Performance

Demand is concentrated: the average category generates 1,344 orders, but **bed_bath_table** alone accounts for 9,417 orders, followed by health_beauty (8,836) and sports_leisure (7,720). Price is essentially decoupled from volume – computer-related categories carry by far the highest average item prices, while most high-volume categories sit close to or below the platform-wide average price of \$120.65. Review scores average 4.09 platform-wide, but several categories sit meaningfully below this, including diapers_and_hygiene, several kitchen-appliance categories, and security_and_services.

Key finding. High order volume, high price, and high satisfaction are three largely independent axes of category performance – no single category dominates on all three.

3.5. Payment Behavior

Credit card is the dominant payment method (73,764 of roughly 99,000 orders) and carries the highest average order value (\$166.60), compared to \$105.25 for voucher-only payments. Order value rises steadily with installment count – median order value grows from \$79 (single payment) to \$248 (13+ installments) – but review scores are essentially flat across payment types (4.05–4.18) and across installment groups.

Key finding. Payment behavior predicts basket size, not satisfaction – installments appear to be a financing choice tied to purchase size rather than a source of customer friction.

3.6. Root Cause Synthesis

Combining the above: **delivery lateness is the primary, clearly non-linear driver of low review scores**; **product category and geographic distance are secondary contributors** of smaller and more diffuse magnitude; and **payment method shows no meaningful association** with satisfaction once basket size is accounted for.

4. Supporting Visualizations

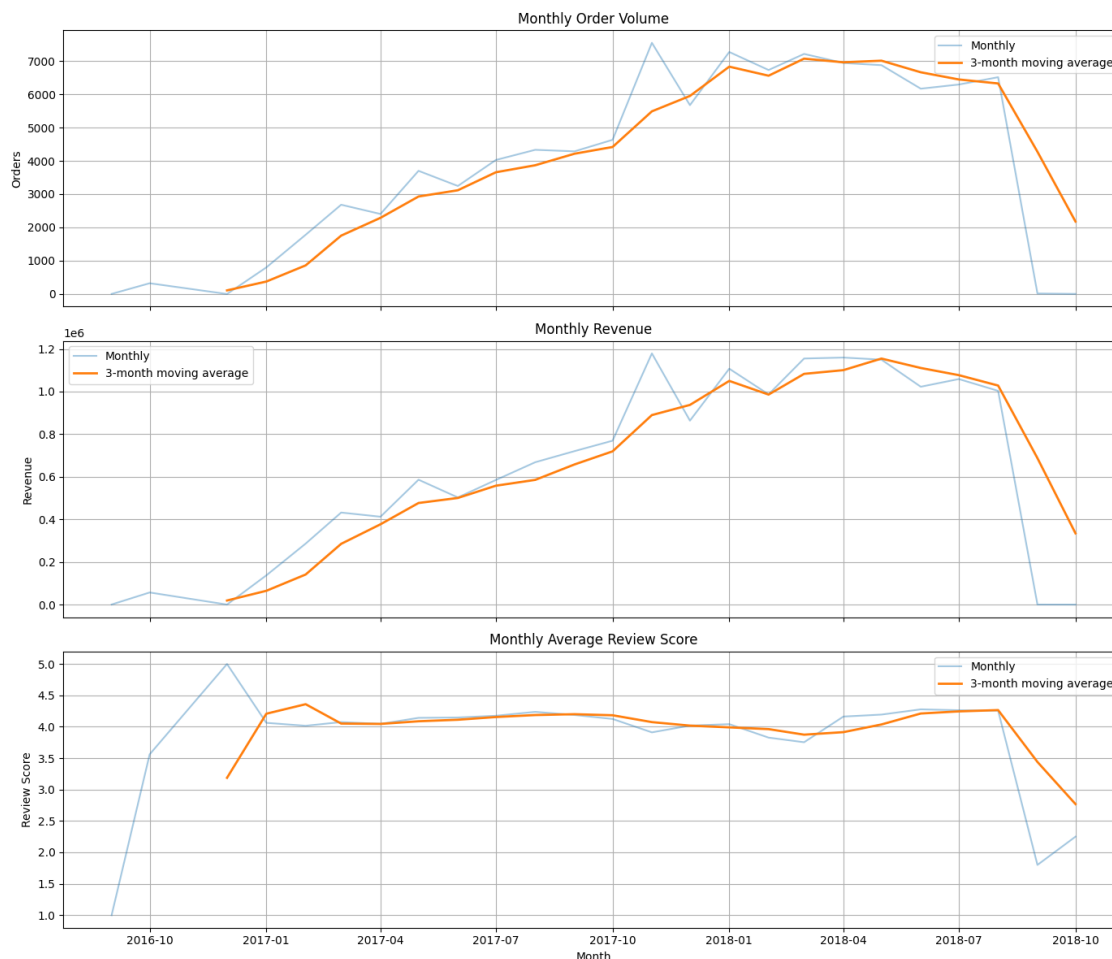


Figure 1: Monthly order volume, revenue, and average review score, with 3-month moving averages. Orders and revenue move together, while review score follows an independent, comparatively stable path around 4.0–4.2. The sharp drop in the final month(s) reflects a well-known data-completeness cutoff in this dataset (the last month of records is incomplete) rather than a genuine collapse in activity, and should be discounted when reading the trend.

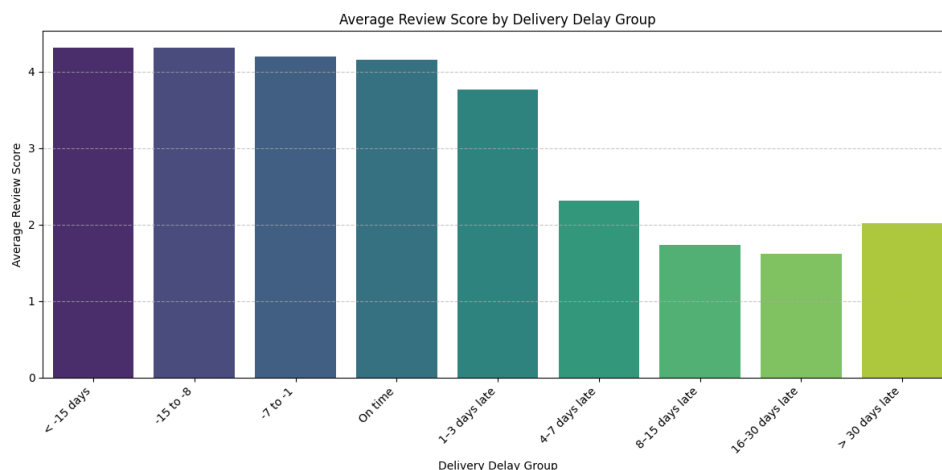


Figure 2: Average review score by delivery-delay group. Scores are essentially flat for on-time and early deliveries, then fall sharply once deliveries run late – illustrating the non-linear, threshold-like nature of the delay–satisfaction relationship referenced in Section 3.2.

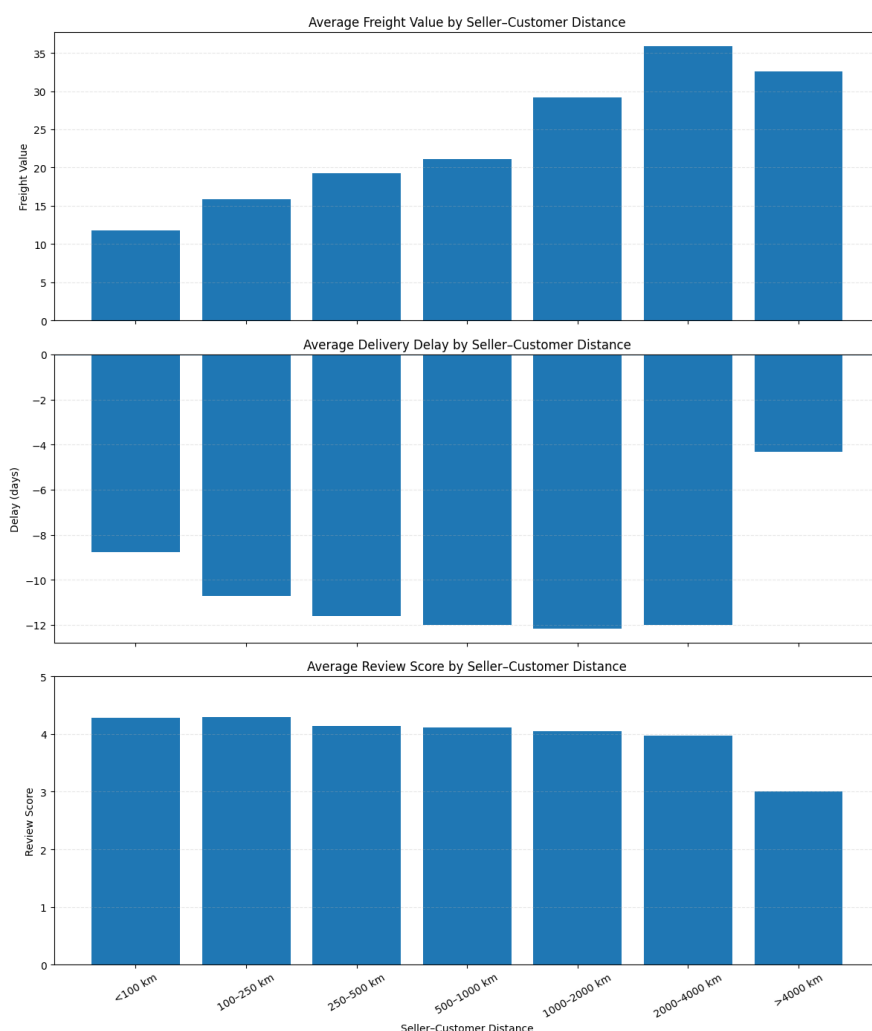


Figure 3: Average freight value, delivery delay, and review score by seller–customer distance band. Freight cost rises monotonically with distance (top panel); delivery delay stays negative (early) at every distance band (middle panel), showing that Olist’s estimated delivery dates already account for distance; review score declines only mildly with distance (bottom panel, note the final “>4000 km” bar rests on just 4 orders).

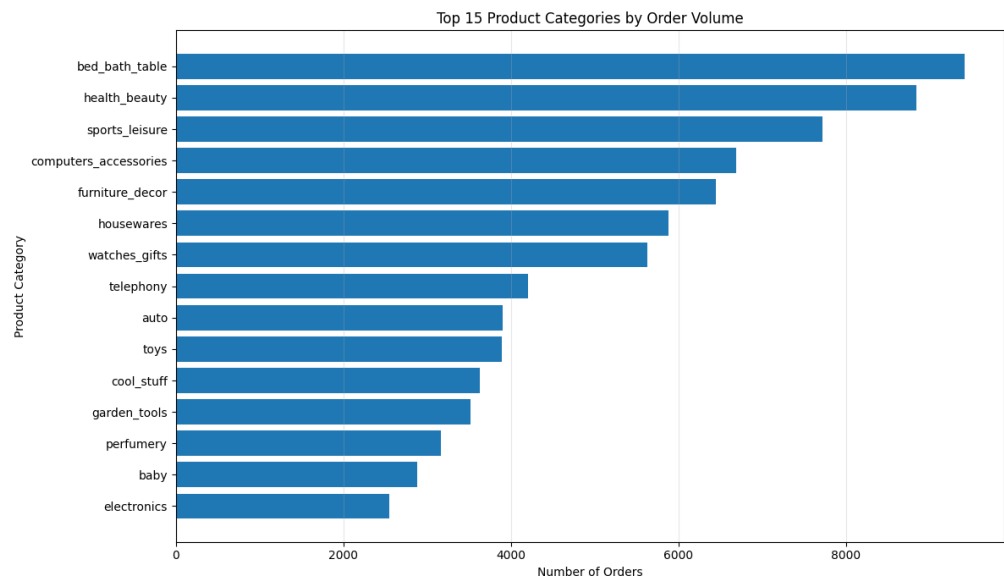


Figure 4: Top 15 product categories by order volume. Demand is concentrated in a small number of categories, led by bed_bath_table, health_beauty, and sports_leisure.

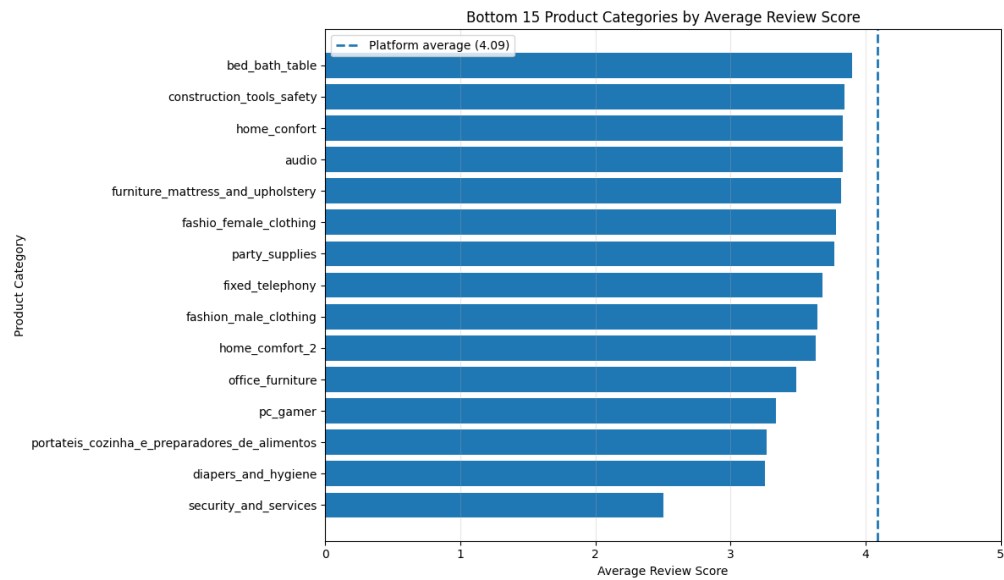


Figure 5: Bottom 15 product categories by average review score, against the platform average (dashed line, 4.09). Several categories sit meaningfully below average; the lowest bar (security_and_services) is based on only 2 orders and should be read as indicative at most, not as a reliable estimate.

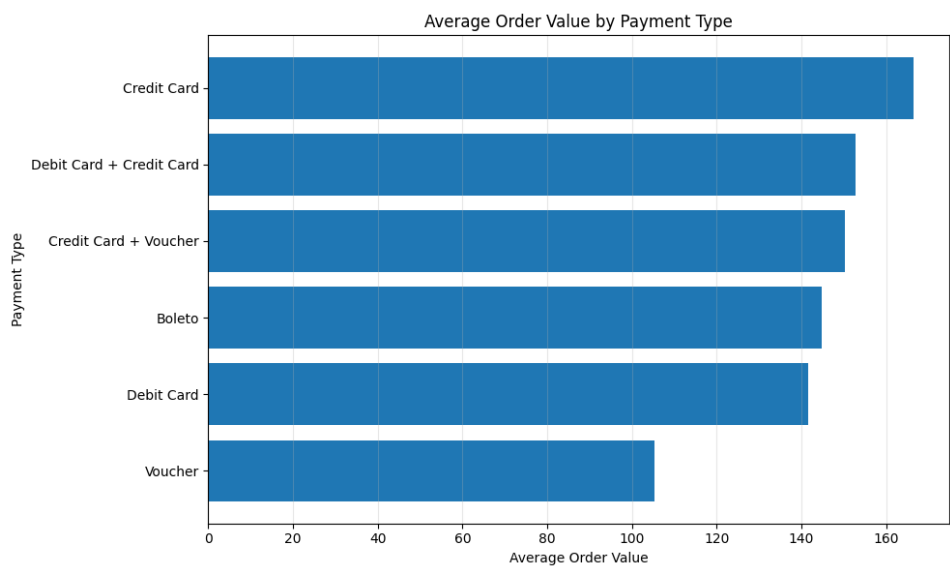


Figure 6: Average order value by payment type. Credit card carries both the highest order value and the largest share of orders; voucher-only payments carry the lowest average value. (Average review score is not shown here as it varies by less than 0.15 points across all payment types – see Table 3.)

5. Business Findings

5.1. Platform Snapshot

Table 1: Platform-level KPIs across the observed period.

Metric	Value
Total orders	99,441
Total revenue	\$15,843,553
Average review score	4.09 / 5
On-time delivery rate	91.9%
Average freight cost per item	\$19.99

Olist's baseline performance is strong: satisfaction sits comfortably above the midpoint of the rating scale, and the large majority of orders (91.9%) arrive at or before the estimated delivery date. This context matters – the findings below concern what moves satisfaction *at the margin*, not a platform in crisis.

5.2. Delivery Timing Is the Dominant Lever

Because the drop in satisfaction is concentrated almost entirely in the minority of orders that run late, the business impact of delivery performance is disproportionate to its frequency: with roughly 8% of orders arriving late, fixing even a fraction of these has an outsized effect on average sentiment compared to broad, platform-wide initiatives.

5.3. Distance Is a Cost Center, Not a Satisfaction Crisis

Table 2: Freight and review score by seller–customer distance (selected bands).

Distance band	Avg. freight (\$)	Avg. delay (days)	Avg. review
<100 km	11.8	−8.8	4.28
250–500 km	19.3	−11.6	4.14
1000–2000 km	29.3	−12.0	4.05
2000–4000 km	36.0	−12.0	3.98

Freight roughly triples from the shortest to the longest reliable distance band, while delivery delay stays negative (early) throughout and review score declines by only about 0.3 points. This is a logistics cost-management story, not a customer-experience emergency.

5.4. Category Performance Diverges Across Three Independent Axes

The categories driving the most orders (bed_bath_table, health_beauty, sports_leisure) are not the highest-priced (computers, appliances) nor systematically the best- or worst-reviewed. This means volume-based prioritization, price-based prioritization, and satisfaction-based prioritization of category investment will each point to a different set of categories.

5.5. Payment Is a Financing Signal, Not an Experience Signal

Average order value varies by over \$60 across payment types, while average review score varies by no more than 0.13 points – payment choice is informative about purchase size and financing preference, but not about the customer experience.

Table 3: Order value and satisfaction by payment type.

Payment type	Orders	Avg. order value (\$)	Avg. review
Credit Card	73,764	166.60	4.10
Boleto (bank slip)	19,614	144.91	4.11
Credit Card + Voucher	2,226	150.27	4.09
Debit Card	1,520	141.52	4.18
Voucher	1,540	105.25	4.05

6. Actionable Recommendations

1. **Build a late-delivery early-warning system.** Since satisfaction craters specifically once an order crosses into "late" territory, flag in-transit orders that are trending toward missing their estimated delivery date and trigger proactive communication, expedited handling, or goodwill compensation before the delivery – and the review – happens.
2. **Set an internal SLA around the 4–7 day late threshold.** The steepest drop in review score occurs once delay exceeds roughly 3–4 days; treat this as the operational red line for carrier escalation rather than waiting for the full 30-day tail.
3. **Preserve (and monitor) the distance buffer already built into estimated delivery dates.** The current estimation logic appears to successfully absorb extra travel time for distant orders (delay stays negative across every distance band). As volume grows, re-validate that this buffer keeps holding for the most remote bands rather than assuming it will scale automatically.
4. **Run targeted quality-control reviews on adequately sampled low-scoring categories,** rather than platform-wide interventions. Restrict action to categories with a meaningful number of orders; categories with single-digit order counts (e.g., security_and_services, n=2) should be tracked but not acted on until more data accumulates.
5. **Treat installment-heavy, high-value orders as a finance and fraud/repayment topic, not a satisfaction topic.** Since installment count tracks order value without affecting review scores, any concerns about long installment plans should be handled through credit and collections policy rather than customer-experience initiatives.
6. **Track a small, recurring KPI dashboard** (order volume, revenue, average review score, on-time rate, average freight) on a monthly cadence, and validate any apparent trend break – including the sharp end-of-period drop visible in Figure 1 – against known data-completeness boundaries before treating it as a real business signal.

7. Conclusion

Across every core question examined, one theme recurs: **delivery timing is the single most important lever connected to customer satisfaction on Olist**, with an effect that is large, concentrated in the late-delivery tail, and consistent across categories and geography. Product category and seller–customer distance are real but secondary contributors – distance in particular affects cost far more than experience, thanks to delivery-date estimation that already compensates for travel time. Payment method and installment behavior are essentially unrelated to satisfaction once basket size is taken into account.

It is worth restating clearly that these are **associations, not proven causal effects**: no experimental or quasi-experimental design (e.g., randomized delivery delays) was used, so a plausible confound – for instance, a product or region that is independently prone to both delays and

quality issues – cannot be fully ruled out from correlational analysis alone. Several results also carry small-sample caveats, most notably the worst-performing product category (2 orders) and the furthest geographic distance band (4 orders), both of which should be treated as indicative rather than reliable. With those caveats acknowledged, the evidence consistently points to delivery reliability – not price, category mix, geography, or payment method – as the highest-leverage, most actionable target for improving customer satisfaction on the platform.